Science - Unit 3 – Practice test

Name:		Class:		Date:	

Unit 3: Materials

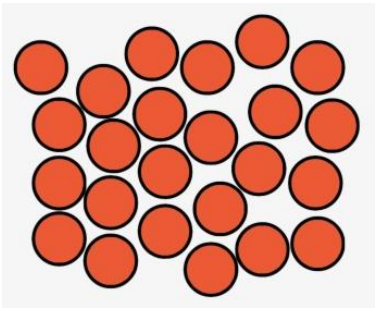
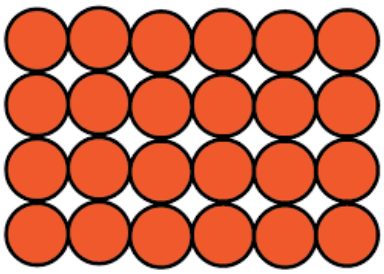
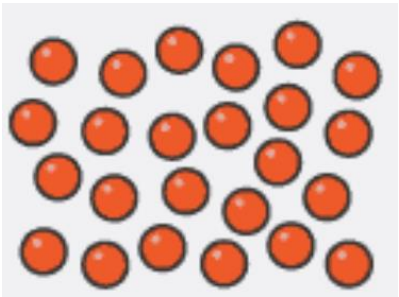
1. Label the types of materials: [4pts]

*	*	*	
*	*	*	
metal	glass	plastic	wood

2. Choose TRUE or FALSE for the statements below: [3pts]

All materials are made of matter and matter is made up of particles.	TRUE FALSE
Particles in solids are packed tightly together and don't move.	TRUE FALSE
Substances cannot change their states.	TRUE FALSE

3. Look at the particle models below and choose which state they represent: [3pts]

		
*	*	*
*	*	*
solid	gas	liquid

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4. Match the beginnings of the sentences with their endings to make true statements: [3pts]

The particles in a solid are held together strongly.	*	*	They have hardly any space between particles so they can not easily move.
The particles in a liquid are separated by larger spaces.	*	*	These spaces allow the particles to move and slide past each other.
Powders are bulk solids composed of many very fine particles.	*	*	They may flow freely and take the shape of a container like a liquid.

5. Look at the picture and answer the following questions: [3pts]

In what state is the chocolate when a person puts it in the pot?	solid liquid gas
In what state will the chocolate be after heating?	solid liquid gas
What is the change of state called?	melting solidifying natural process chemical reaction

6. Match the things with what they are: [4pts]

salt	*	*	liquid
oxygen	*	*	gas
glass	*	*	solid
water	*	*	material

7. Choose the correct word to complete the change of state processes: [2pts]

Solid + **heating** / **cooling** = liquid
Liquid + **heating** / **cooling** = solid

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8. Choose the best answer for each of the questions below: [2pts]

		*	Its particles slow down and move closer together.
What happens when the liquid loses heat energy and cools?	*	*	Its particles move faster and get further apart.
		*	Its particles keep moving at the same speed.

		*	Its particles move faster.
What happens when the solid gets heat energy?	*	*	Its particles move slower.
		*	Its particles keep moving at the same speed.

9. Look at the pictures and decide if chemical reactions have taken place or not: [2pts]

chemical reaction		no chemical reaction

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chemical reaction		no chemical reaction

10. Match the substance with the materials used in the chemical reactions: [4pts]

Cement powder, sand, water + limestone	*	*	ash and gas
Wood + fire	*	*	rust
Metal, water + air	*	*	concrete
Sand, soda ash, limestone + heat	*	*	glass